Counterterrorism Spending and Terrorist Attack Severity

A country-year quantitative study of military expenditure and terrorism outcomes, 2001-2020

Prepared for portfolio and LinkedIn document upload

Key Takeaways
The final model dataset includes 1,382 country-year observations.
Basic year fixed-effects models show a positive and statistically significant association between lagged military spending and terrorism outcomes.
After adding country fixed effects, the spending coefficient becomes small and statistically insignificant for both attacks and deaths.
The results do not support the claim that higher military spending independently reduces terrorism; the basic positive relationship likely reflects cross-country differences in terrorism risk.

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Generated from Python analysis outputs and research documentation.

Introduction

This project examines the relationship between terrorism outcomes and military/security spending using a country-year panel dataset from 2001 to 2020. The central research question is whether higher prior-year military spending is associated with fewer terrorist attacks or fewer terrorism-related deaths.

The study uses military expenditure as a proxy for security and counterterrorism-related spending. This proxy is imperfect because military expenditure includes broad defense spending and does not isolate spending specifically directed toward counterterrorism, intelligence, policing, border security, or prevention programs. For that reason, the findings should be interpreted as evidence about the relationship between military spending and terrorism outcomes, not as a direct causal estimate of counterterrorism program effectiveness.

Data

The terrorism data come from the Global Terrorism Database and are aggregated from incident-level observations into country-year totals. The main terrorism outcomes are the number of terrorist attacks, terrorism deaths, wounded victims, casualties, and fatalities per attack. Military expenditure, GDP, and population data are taken from World Bank/SIPRI indicators and merged into the country-year panel.

The final model dataset includes 1,382 country-year observations.

Policy Context: Why Military Spending Is Only a Proxy

A central challenge in this project is that counterterrorism spending is not reported as one clean, globally comparable category. Military expenditure is available across many countries and years, which makes it useful for a broad country-year study, but counterterrorism policy extends far beyond military budgets alone.

The U.S. Department of State's Bureau of Counterterrorism describes modern counterterrorism as involving diplomacy, multilateral cooperation, civilian capacity-building, border and aviation security, sanctions and terrorist designations, countering violent extremism, and partner support. This shows that counterterrorism is partly military, but also institutional, diplomatic, financial, legal, and law-enforcement oriented.

Brown University's Costs of War project also demonstrates the breadth of counterterrorism activity. Its report on U.S. counterterrorism operations from 2021 to 2023 documents activity in 78 countries, including training and assistance in 73 countries, ground combat in at least nine countries, and air or drone strikes in four to five countries. This supports the interpretation that military expenditure is a broad proxy rather than a precise measure of counterterrorism spending.

Variables

Dependent Variables

- Terrorist attacks per country-year
- Terrorism deaths per country-year
- Log terrorist attacks

- Log terrorism deaths

Main Independent Variable

- Lagged log military spending, measured as prior-year military expenditure. The lagged variable is used to reduce same-year reverse-causality concerns.

Control Variables

- GDP
- Population
- Year fixed effects
- Country fixed effects in the stronger panel models

Descriptive Findings

The descriptive charts show that global terrorist attacks and terrorism deaths rose sharply after 2011, peaked around 2014, and then declined through the end of the study period. Terrorism was also highly concentrated geographically. Iraq and Afghanistan were the most affected countries by total attacks, while Iraq, Afghanistan, Nigeria, Pakistan, and Syria were among the countries with the highest terrorism-related death totals.

The scatterplots comparing military spending and terrorism outcomes do not show a simple negative relationship. In other words, countries with higher military spending do not clearly experience fewer attacks or fewer deaths in the raw visual patterns. This supports the need for regression models that control for GDP, population, year effects, and country-level differences.

Selected Charts

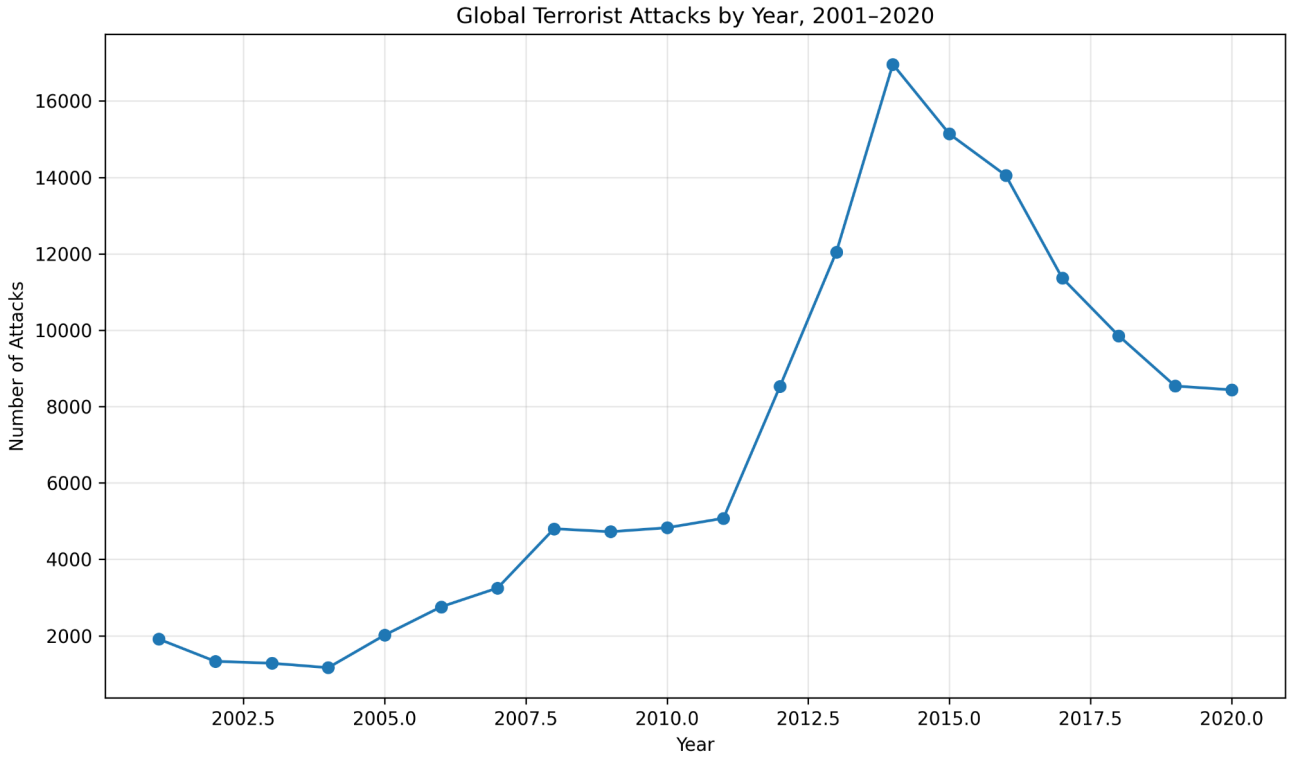


Figure 1. Global terrorist attacks by year, 2001-2020.

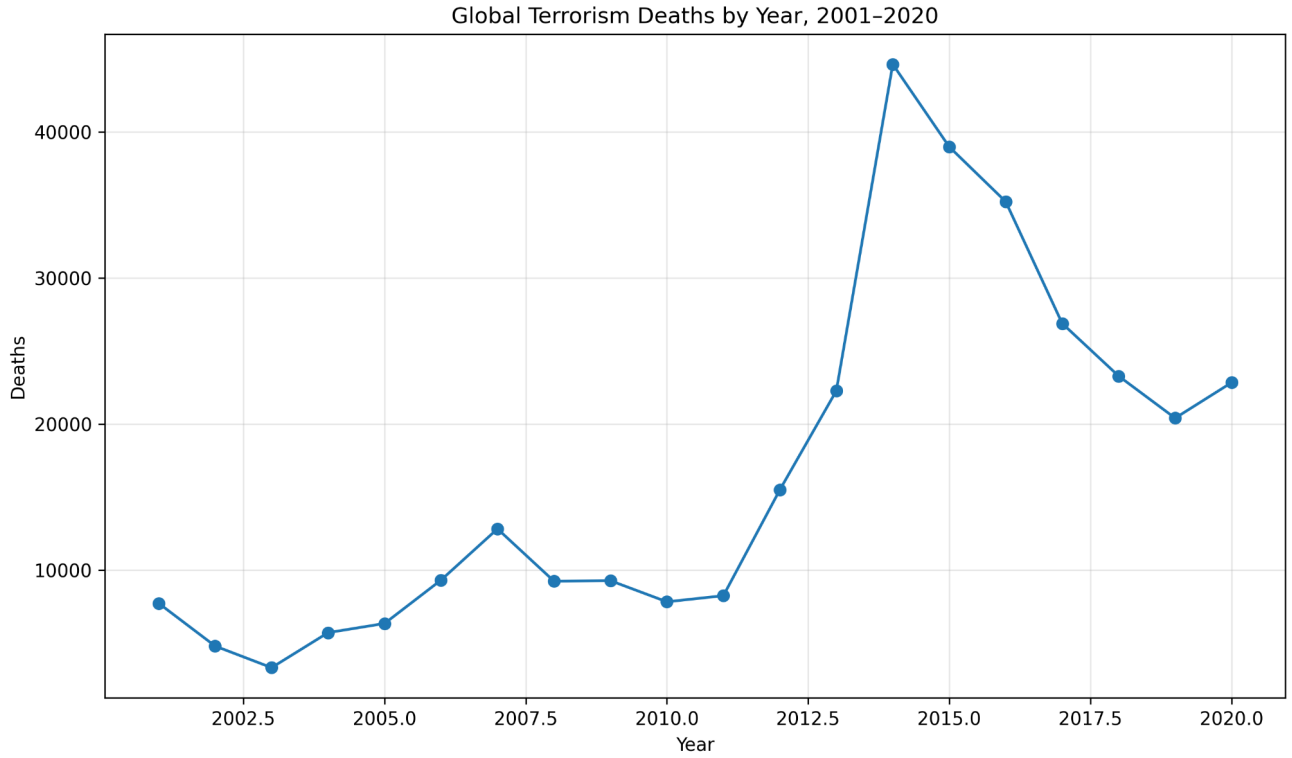


Figure 2. Global terrorism deaths by year, 2001-2020.

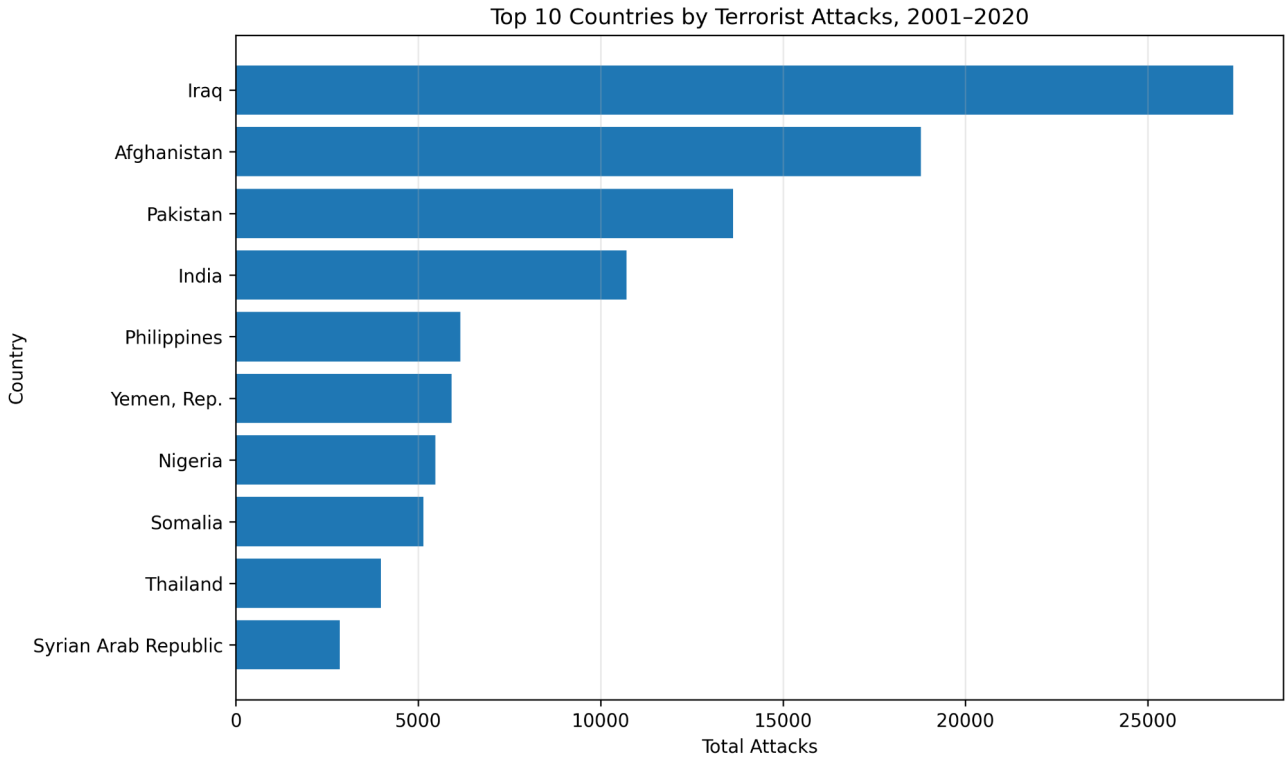


Figure 3. Top 10 countries by total terrorist attacks, 2001-2020.

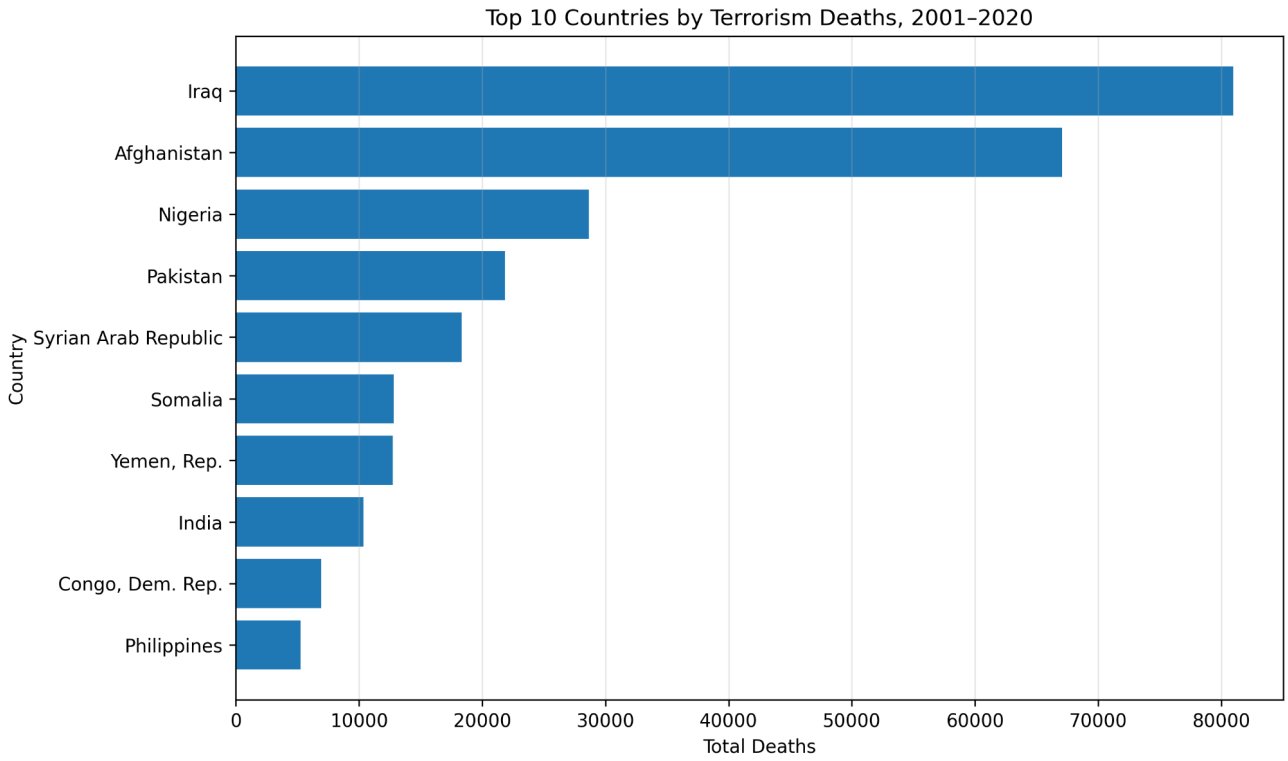


Figure 4. Top 10 countries by terrorism deaths, 2001-2020.

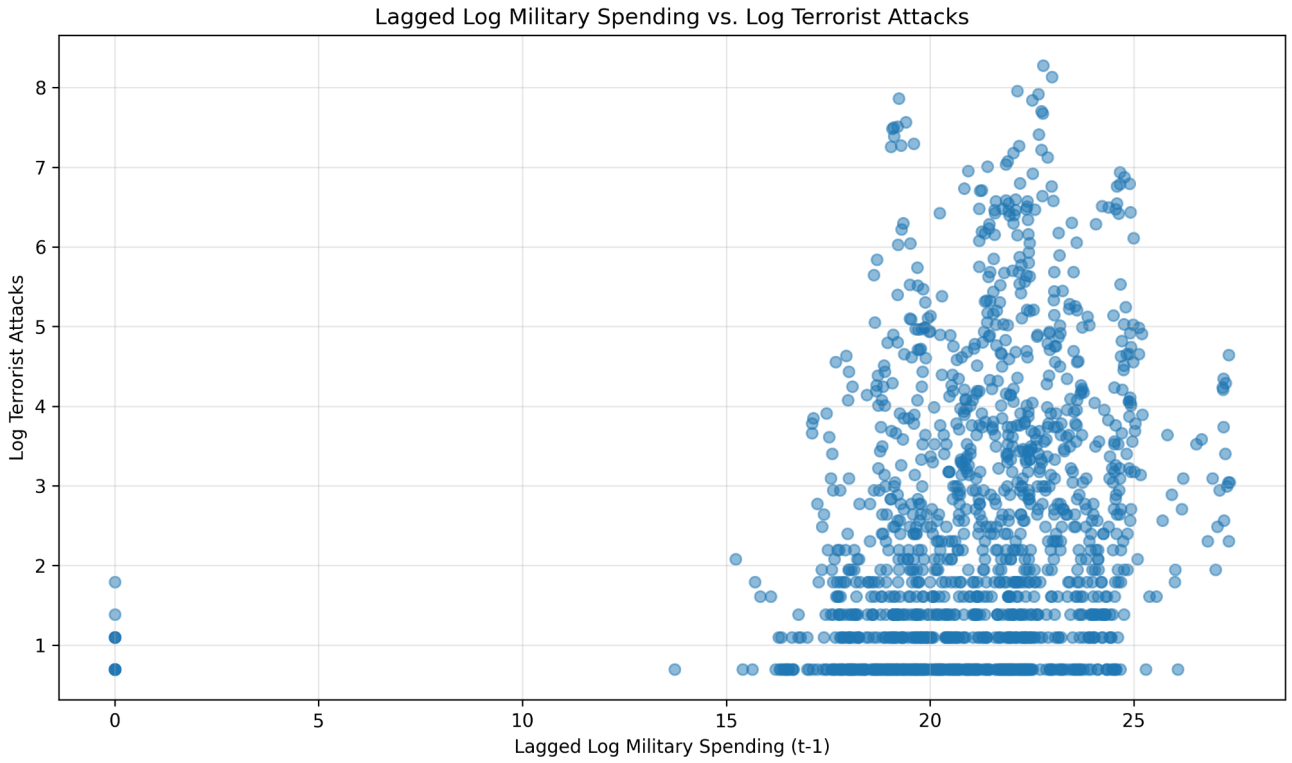


Figure 5. Lagged log military spending versus log terrorist attacks.

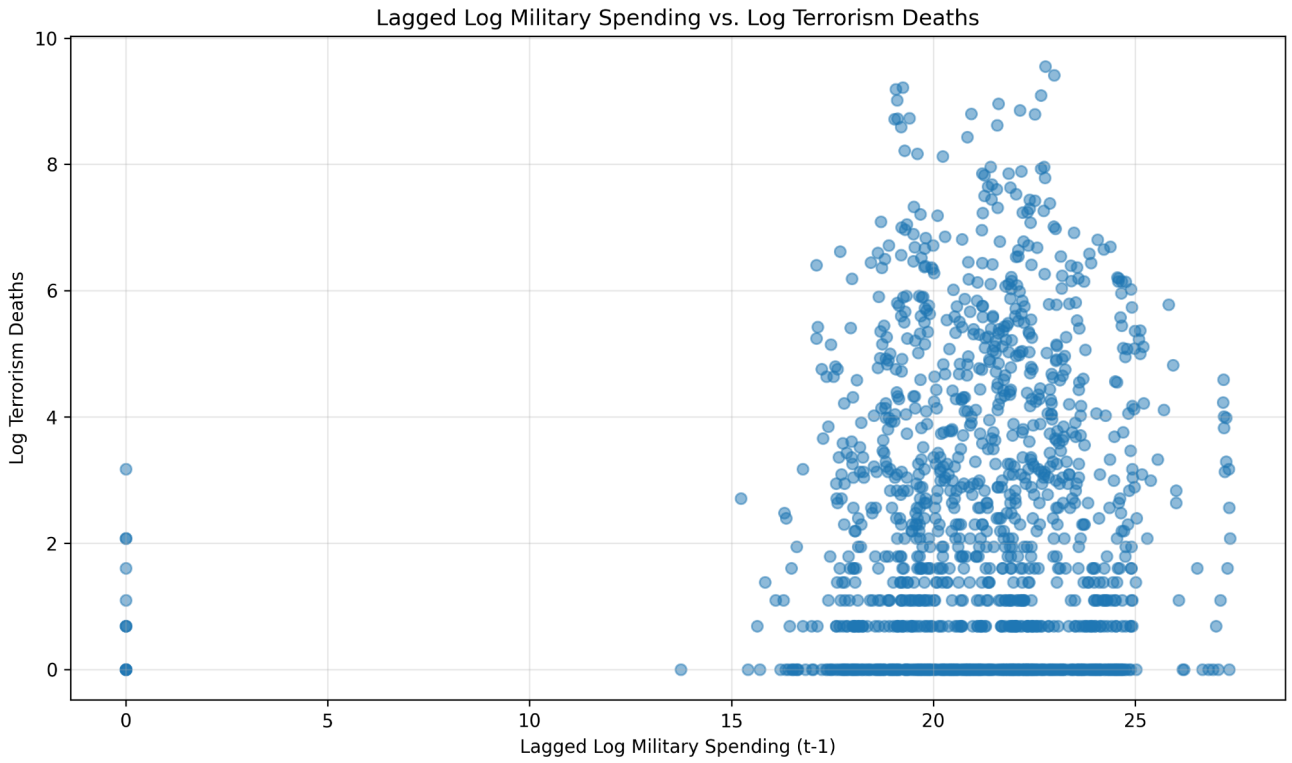


Figure 6. Lagged log military spending versus log terrorism deaths.

Regression Approach

The regression analysis compares two types of models. The first set includes year fixed effects only. These models compare countries to one another while controlling for global year-specific shocks. The second set includes both country and year fixed effects. These stronger panel models compare each country more closely against itself over time, reducing bias from stable differences between countries.

Regression Findings

The regression results show that model specification strongly affects the relationship between military spending and terrorism outcomes. In the basic models with year fixed effects only, lagged military spending is positively and statistically significantly associated with both terrorist attacks and terrorism deaths. However, after adding country fixed effects, the coefficient on lagged military spending becomes small and statistically insignificant.

Model	Outcome	Coefficient	Std. Error	P-value	Significant?
Model 1A - Year FE attacks model	Log terrorist attacks	0.1043	0.0208	5.361e-07	Yes
Model 1B - Year FE deaths model	Log terrorism deaths	0.1196	0.0269	8.507e-06	Yes
Model 2A - Country and Year FE attacks model	Log terrorist attacks	-0.0026	0.0265	0.9232	No
Model 2B - Country and Year FE deaths model	Log terrorism deaths	-0.0301	0.0451	0.5039	No

This pattern suggests that the positive relationship in the basic models is likely driven by cross-country differences. Countries facing greater terrorism risk also tend to spend more on defense and security. Once countries are compared more closely against their own historical patterns, lagged military spending is not a statistically significant predictor of later terrorism outcomes.

Discussion

The most important finding is that the relationship between military spending and terrorism outcomes changes substantially after country fixed effects are included. In the basic models, lagged military spending is positively and statistically significantly associated with terrorist attacks and terrorism deaths. However, once country fixed effects are added, the coefficient becomes small and statistically insignificant for both outcomes.

This suggests that the positive relationship in the basic models is likely driven by differences between countries. Countries facing greater terrorism risk tend to spend more on defense and security, so cross-country comparisons can make spending appear positively associated with terrorism. When countries are compared more closely against their own historical patterns, military spending does not appear to be a statistically significant predictor of later terrorism outcomes.

Limitations

There are several important limitations. First, military expenditure is an imperfect proxy for counterterrorism spending. It includes broad defense spending and does not isolate spending specifically directed toward terrorism prevention, intelligence, law enforcement, border security, homeland security, sanctions enforcement, foreign assistance, or civilian capacity-building.

Second, the study does not fully measure domestic law enforcement, intelligence, homeland security, or foreign aid programs aimed at counterterrorism. Third, the analysis is observational, so it cannot prove causation. Fourth, terrorism data may be affected by reporting differences across countries and years. Finally, countries often increase security spending in response to terrorism threats, making reverse causality a central concern.

These limitations are especially important because recent counterterrorism policy documents show that counterterrorism activity includes a wide range of tools beyond military operations. State Department reporting emphasizes civilian capacity-building, border security, diplomacy, multilateral coordination, and partner support, while Costs of War research documents global U.S. counterterrorism activity across training, assistance, combat, and air or drone operations.

Future Research

Future research could improve on this study by using more detailed counterterrorism-specific spending measures. Rather than relying only on aggregate military expenditure, a future project could separate defense spending, homeland security spending, foreign assistance, border security programs, intelligence-related spending, counterterrorism training, sanctions enforcement, and civilian capacity-building.

A U.S.-focused extension could also use sources such as USAspending.gov, Department of Defense budget documents, Department of Homeland Security budget documents, State Department counterterrorism foreign assistance data, and Costs of War estimates to create a more detailed spending dataset. This would allow researchers to examine whether some types of counterterrorism spending are more closely associated with changes in terrorist attacks and terrorism-related deaths than others.

Conclusion

Overall, the results do not provide evidence that higher military spending reduces terrorist attacks or terrorism deaths. The basic models show a positive association between lagged military spending and terrorism outcomes, but this relationship disappears in the stronger country fixed effects models. The findings therefore suggest that cross-country differences in terrorism risk explain much of the apparent relationship between spending and terrorism. A more precise study of counterterrorism effectiveness would require narrower data on actual counterterrorism, intelligence, policing, border security, foreign assistance, civilian capacity-building, and prevention expenditures.

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